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MATHEMATICS

STANDARD FOUR

National Assessment 2023
Answer Key and Worked Solutions

Subject	Mathematics
Code	04E
Level	Standard Four
Year	2023
Exam Board	NECTA
Exam	National Assessment
Type	Answer key and worked solutions
Questions	5 questions, each with 5 parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S4-2023.pdf

Question 1: Numbers and Place Value

1(a). Write XLII in Arabic numerals.

Step 1: XL means $50 - 10 = 40$.

Step 2: II = 2; $40 + 2 = 42$.

Final Answer: 42.

1(b). Write the football team's age, 22 years, in Roman numerals.

Step 1: $20 = XX$; $2 = II$.

Step 2: Join the symbols.

Final Answer: XXII.

1(c). Write 2020 in words.

Step 1: $2,000 =$ two thousand.

Step 2: $20 =$ twenty.

Final Answer: Two thousand and twenty.

1(d). Form a number with ten-thousands 1, thousands 7, hundreds 4, tens 1 and ones 3.

Step 1: Write the digits in their specified places: 1, 7, 4, 1, 3.

Step 2: Their values total $10,000 + 7,000 + 400 + 10 + 3$.

Final Answer: 17,413.

1(e). Write nine thousand and fifty shillings in figures.

Step 1: Nine thousand = 9,000; fifty = 50.

Step 2: $9,000 + 50 = 9,050$.

Final Answer: Sh. 9,050.

Section Summary: Read each digit's place and distinguish ordinary numbers from Roman numerals.

Question 2: Sequences and Comparison

2(a). Find the missing number: 90, 85, 80, 75, __, 65.

Step 1: Each number decreases by 5.

Step 2: $75 - 5 = 70$; $70 - 5 = 65$ checks.

Final Answer: 70.

2(b). Which is greater: 84×6 or 86×4 ?

Step 1: $84 \times 6 = 504$.

Step 2: $86 \times 4 = 344$.

Step 3: Compare 504 and 344.

Final Answer: 84×6 is greater ($504 > 344$).

2(c). Find the missing fraction: $\frac{1}{2}$, $\frac{1}{4}$, __, $\frac{1}{16}$.

Step 1: The denominator doubles: 2, 4, 8, 16.

Step 2: The numerator stays 1.

Final Answer: $\frac{1}{8}$.

2(d). A patient weighs 90 kg in the first month and loses 1 kg each month. Find the sixth month's weight.

Step 1: From month 1 to month 6, there are five losses.

Step 2: $90 - 5 \times 1 = 85$.

Final Answer: 85 kg.

2(e). Identify the operation in 99, 89, 79, 69, 59, 49, 39.

Step 1: Compare adjacent terms: $99 - 89 = 10$, $89 - 79 = 10$.

Step 2: Each term is found by subtracting ten from the previous one.

Final Answer: Subtract 10 each time.

Section Summary: Find the repeated change first; count the gaps between terms.

Question 3: Operations and Time

3(a). A book has 91 pages with 13 pictures each. Find the total pictures.

Step 1: Multiply pages by pictures per page.

Step 2: $91 \times 13 = 91 \times 10 + 91 \times 3 = 910 + 273$.

Final Answer: 1,183 pictures.

3(b). Calculate 36×8 (the expression confirmed in the Kiswahili paper).

Step 1: $36 = 30 + 6$.

Step 2: $30 \times 8 = 240$; $6 \times 8 = 48$.

Step 3: $240 + 48 = 288$.

Final Answer: 288.

3(c). Add 14 hours 30 minutes and 8 hours 40 minutes.

Step 1: $30 + 40 = 70$ minutes = 1 hour 10 minutes.

Step 2: $14 + 8 + 1 = 23$ hours.

Final Answer: 23 hours 10 minutes.

3(d). Convert 3 hours 45 minutes into minutes.

Step 1: $3 \times 60 = 180$ minutes.

Step 2: $180 + 45 = 225$.

Final Answer: 225 minutes.

3(e). Subtract sh. 1,300 from sh. 1,980.

Step 1: $1,980 - 1,000 = 980$.

Step 2: $980 - 300 = 680$.

Final Answer: Sh. 680.

Section Summary: Keep the units consistent. Regroup every 60 minutes as one hour.

Question 4: Shapes and Measurement

4(a). Find the perimeter of a rectangle 100 cm long and 50 cm wide.

Step 1: $P = 2 \times (\text{length} + \text{width})$.

Step 2: $2 \times (100 + 50) = 2 \times 150$.

Final Answer: 300 cm.

4(b). Name the shape with vertices A, B and C.

Step 1: The figure has three straight sides and three corners.

Step 2: A shape with three sides is a triangle.

Final Answer: A triangle.

4(c). A rectangle has perimeter 80 cm and length 25 cm. Find its width.

Step 1: $\text{Length} + \text{width} = \text{perimeter} / 2 = 40 \text{ cm}$.

Step 2: $\text{Width} = 40 - 25$.

Final Answer: 15 cm.

4(d). A rectangular football ground has perimeter 480 m and width 40 m. Find its length.

Step 1: $\text{Length} + \text{width} = 480 / 2 = 240 \text{ m}$.

Step 2: $\text{Length} = 240 - 40$.

Final Answer: 200 m.

4(e). A square has perimeter 60 cm. Find one side.

Step 1: All four sides are equal.

Step 2: $\text{Side} = 60 / 4$.

Final Answer: 15 cm.

Section Summary: Use the shape's sides, perimeter formula and metric units carefully.

Question 5: Reading Data

5(a). The harvest pictograph has 3 sacks in 2015, 6 in 2016 and 4 in 2017. Which year has the most?

Step 1: Count each row carefully.

Step 2: Six sacks is the largest count.

Final Answer: 2016.

5(b). One sack represents 1,000 kg. Find the 2017 harvest.

Step 1: There are four sacks in the 2017 row.

Step 2: $4 \times 1,000 = 4,000$.

Final Answer: 4,000 kg.

Study Tip: Count the symbols in the source, rather than relying on a previous key.

5(c). Which year has the smallest harvest?

Step 1: Compare 3, 6 and 4 sacks.

Step 2: Three sacks is the smallest count and is shown for 2015.

Final Answer: 2015.

5(d). Find the difference in harvest between 2015 and 2017.

Step 1: 2015: $3 \times 1,000 = 3,000$ kg; 2017: $4 \times 1,000 = 4,000$ kg.

Step 2: $4,000 - 3,000 = 1,000$.

Final Answer: 1,000 kg.

5(e). Which year had a 6,000 kg harvest?

Step 1: $6,000 / 1,000 = 6$ sacks.

Step 2: The six-sack row is 2016.

Final Answer: 2016.

Section Summary: Read the key, count every symbol, and distinguish the year from the numerical value.